0580_w24_ms_43_Q7

Question	Answer	Marks	Partial Marks	Guidance
7(a)(i)	88	**2**	**M1** for $\frac{1}{2}(9+13)\times 8$ oe	
7(a)(ii)(a)	$\frac{1}{2}(y+4+y+1)\times(y+2)$ [=264] or $\frac{1}{2}\times 3\times(y+2)+(y+1)\times(y+2)$ [=264]	**M1**		Allow $2y + 5$ for sum of sides For M1 and B1 allow recovery from omission of brackets if expansion is correct but then A0. For M1 allow $(y + 2.5)(y + 2)$
	$2y^2+5y+4y+10$	**B1**		Allow $9y$ for $5y + 4y$ Allow $y^2 + 2.5y + 2y + 5$
	Leading to $2y^2+9y-518=0$	**A1**	No errors or omissions	
7(a)(ii)(b)	$(2y+37)(y-14)$	**B2**	**B1** for $(2y+a)(y+b)$ where $ab = -518$ or $a + 2b = 9$ or $2y(y-14)+37(y-14)$ or $y(2y+37)-14(2y+37)$	Allow B2 for eg $2y(y - 14) + 37(y - 14)$ [= 0] followed by a correct solution. Award B0 for finding roots and working back as $(y + 18.5)(y - 14)$ but allow B2 if roots found first and then final answer is $(2y+37)(y-14)$. Condone omission of final bracket for B2 or B1.
	14	**B1**		Condone inclusion of -18.5 oe.

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Question	Answer	Marks	Partial Marks	Guidance
7(b)	26.5 or 26.47...	**3**	**B2** for 10.5 or 10.47… or $\frac{10\pi}{3}$ OR **M2** for $8 + 8 + \frac{75}{360} \times 2\pi 8$ or **M1** for $\frac{75}{360} \times 2\pi 8$	Allow use of $\frac{22}{7}$ or 3.14 for all M marks.
7(c)	$25 + \frac{25}{2}\pi$	**4**	**M2** for $\frac{90}{360} \times \pi \times (\sqrt{(5^2 + 5^2)})^2$ or **M1** for [radius ^{2} =] $5^2 + 5^2$ **M1** for [triangle area =] [2×] $\frac{1}{2} \times 5 \times 5$ oe	Accept $25 + 12.5\pi$ **M2 M1** implied by 64.3 or 64.26 to 64.28. For M2 allow radius 7.07[1…] or $5\sqrt{2}$. For M2 allow radius = 7 or 7.1 provided $5^2 + 5^2$ seen or $\frac{5}{\sin 45}$ oe seen First M1 implied by $\sqrt{50}$ or 7.07… Second M1 implied by 12.5, oe